SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

Total Marks:40

QUESTIONS: 4

Annexure A

CASE STUDY

Code of Conduct:

(192424426L)
I DILIP.S.N. *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.V.

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NAME : DILIP. S.V.
REG : 192424426.

CO3 - AT - 1

Smart medical Diagnosis

a) propositional Logic.

Let F = Fever, C = cough, R = Rash, B = Breathless

* $F \wedge C \rightarrow$ flu.

* $F \wedge R \rightarrow$ measles.

* $C \wedge B \rightarrow$ Pneumania

Facts

F = True, C = True, B = True.

b) Froward chanining.

* Fever + caugh → Possible flu.

* Fever + Rash → cannot conclude measles.

* cough + Breathlesshess → Possible Pnoumania

conclusion:- Patient A may have Flu, and Pneumonia

c) Backward chanining.

Goal : Pnenmonia.

Rule:

Cough (A ∧) Breathlessness (A) → Pneumonia (A)

Both fact are true.

Therefore.

Pneumonia (A)

2) Question 2 - Autonomous Traffic management

a) unification.

Rule:

Emergency Type (x) → clear path (x)

Fact:

Emergency type (Ambulance)

Substitution:

x / Ambulance;

Therefore.

clear path (Ambulance)

Then:

Green signal (Ambulance)

using Rule: 3.

· Vehicle (CarA) ∧ Behind (CarA, Ambulance) ∧ Green

) Resolution.

Negate the goal

proceed (car A)

using Rule 3 and the given fact, Resolution produces

the empty clause.

Hence:

proceed (car A) is true.

c) Two Emergency vechicles.

The KB is not sufficient. Additional rules

for priority, collision avoidance and scheduling

are required.

Agricultural Expert. System.

a) CNF.

P1:

SoilDry ∨ Irrigation needed.

P2: Irrigtion needed ∨ ¬ corp wheat. ∨ Apply Drip

P3:

Apply Drip method ∨ ¬ crop wheat ∨ crop At Risk.

Fact.

Soil Dry crop wheat.

b) Resolution.

From

Soil Dry.

and P1

Irrigation needed

using P2.

Apply Drip method.

Thus.

Apply Drip method = True.

c) Removing soilDry.

without SoilDry.

Irrigation needed.

cannot be derived.

There for Apply Drip method cannot be driven

) wumpus world.

a) Belief state.

Given.

* Stench [1,2]

* ~~B~~ Breeze [1,1]

* Glitter [2,2]

There fore.

Possible wumpus location.

[1,1], [2,2], [1,3]

Possible wampus locations.

[1,2], [2,1]

Gold location.

[2,2]

b) Forward chaining.

* Stench [1,2] - wumpus is adjacent

* Breeze [1,1] -> pit is adjate.

* Glitter [2,2] -> Gold is at [2,2]

c) Safety of path.

Path.

[1,1] -> [1,2] -> [2,2]

The path is not proven safe, because the KB does not establish that [1,2] and [2,2] are free from pit / wampus.